

MCU / radio (nRF52840 high-voltage mode)

Charging input & charger

Charge inhibit / status

Battery + protection (PCM)

IMU (accelerometer + gyroscope)

Mechanical

Programming & status LED

Power flags (ERC)

design.py – ShotPuck circuit, single source of truth.

gen_sch.py turns this into shotpuck.kicad_sch and gen_pcb.py into shotpuck.kicad_pcb. Change the circuit here, then re-run both.

Power path
J1 Samzo PR5L4015 magnetic receptacle (VIN_RAW), D3 SMF5.0CA bidirectional TVS to GND (ESD on the exposed contacts, hot-plug; bidirectional so a reversed cable is only blocked)
-> D1 BAT54J (reverse polarity) -> VIN
VIN -> R9 1k -> VBUS_SNS (C9 100n) -> U1 VBUS: charger detect only (USB unused, ~24 uA). the RC keeps hot-plug spikes off the 5.8 V abs-max VBUS pin
VIN -> U3 MCP73831-2 (4.20 V, 21 mA via R1 = 47k) -> VBAT (LIR1254, 45-65 mAh)
VBAT -> U1 VDDH (nRF52840 high-voltage mode, REG0 -> VDD 3.0 V)
VDD -> U2 LSM6DS032 (VDD + VDDIO), LED, SWD VCC sense

IMU
U2 LSM6DS032 on I2C at 0x6A (SA0 = GND, CS = VDD selects I2C). Aux master pins SDx/SCx to GND; INT2, OCS_Aux, SDO_Aux unconnected.
Charge inhibit (temperature, firmware)
PROG -> R1 -> Q1A drain; Q1A gate pulled to VIN by R2 (charging allowed whenever a cable is present). Q1B (gate = CHG_INH, R3 pull-down) pulls Q1A gate low -> PROG floats -> MCP73831 charge disabled.

Cell protection (PCM; LIR cells have no protection of their own)
U4 BQ29700 + Q2 DMN2004DWK in the cell's negative lead: BAT_N -> Q2A (DOUT, discharge) -> PCM_D -> Q2B (COUT, charge) -> GND, OVP 4.275 V, UVP 2.80 V, OCD/OCC +/-100 mV across ~1 ohm of FETs (-0.1 A), SCD 0.5 V, R7 330R + C8 100n filter the BAT supply, R8 2k2 on V-. Standby 4 uA.

Current measurement
JP1 (bridged solder jumper) in series with the cell; cut, insert an ammeter, re-bridge with solder.

Charge status
STAT -> D2 (cathode) ; D2 anode -> CHG_STAT GPIO with pull-up enabled only while VBUS is present (no standby leakage).

Generated by hardware/gen_sch.py from design.py

Sheet: /
File: shotpuck.kicad_sch

Title: ShotPuck – arrow shot counter

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/1